

Gelbrich's inequality and Gaussian-preserving Wasserstein flows

Gelbrich's inequality gives a sharp lower bound on the quadratic Wasserstein distance in terms of only means and covariance matrices. It was proved by Gelbrich in [1]. If

$$m_\mu := \int_{\mathbb{R}^d} x \, d\mu(x), \quad \Sigma_\mu := \int_{\mathbb{R}^d} (x - m_\mu)(x - m_\mu)^\top \, d\mu(x),$$

then, for $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$,

$$W_2^2(\mu, \nu) \geq \|m_\mu - m_\nu\|^2 + \text{Tr } \Sigma_\mu + \text{Tr } \Sigma_\nu - 2 \text{Tr} \left((\Sigma_\mu^{1/2} \Sigma_\nu \Sigma_\mu^{1/2})^{1/2} \right).$$

The right-hand side is precisely the squared W_2 distance between the Gaussian laws with the same means and covariances as μ and ν .

1 Definitions

Let $\mathcal{P}_2(\mathbb{R}^d)$ be the space of Borel probability measures on $\mathbb{R}^d$ with finite second moment. For $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$,

$$W_2^2(\mu, \nu) := \inf_{\pi \in \Pi(\mu, \nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|^2 \, d\pi(x, y),$$

where $\Pi(\mu, \nu)$ denotes the set of couplings of μ and ν .

For $A, B \in \mathbb{S}_+^d$, define

$$\mathfrak{B}^2(A, B) := \text{Tr } A + \text{Tr } B - 2 \text{Tr} \left((A^{1/2} B A^{1/2})^{1/2} \right).$$

This is the squared Bures distance between covariance matrices.

Let

$$\mathcal{G}_d := \{N(m, A) : m \in \mathbb{R}^d, A \in \mathbb{S}_+^d\}$$

be the set of possibly degenerate Gaussian laws. We write

$$R\mu := N(m_\mu, \Sigma_\mu)$$

for the Gaussian law with the same mean and covariance as μ .

2 Gelbrich's inequality

Theorem 1 (Gelbrich's inequality). *Let $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$. Then*

$$W_2^2(\mu, \nu) \geq \|m_\mu - m_\nu\|^2 + \mathfrak{B}^2(\Sigma_\mu, \Sigma_\nu).$$

Equivalently,

$$W_2^2(\mu, \nu) \geq \|m_\mu - m_\nu\|^2 + \text{Tr } \Sigma_\mu + \text{Tr } \Sigma_\nu - 2 \text{Tr} \left((\Sigma_\mu^{1/2} \Sigma_\nu \Sigma_\mu^{1/2})^{1/2} \right).$$

Proof. We first prove the matrix estimate needed below.

Lemma 1. *Let X, Y be centered square-integrable $\mathbb{R}^d$ -valued random variables with*

$$\text{Cov}(X) = A, \quad \text{Cov}(Y) = B,$$

where $A, B \in \mathbb{S}_+^d$. Then

$$\text{Tr } \mathbb{E}[XY^\top] \leq \text{Tr} \left((A^{1/2} B A^{1/2})^{1/2} \right).$$

Proof. Set

$$C := \mathbb{E}[XY^\top].$$

The block covariance matrix

$$\begin{pmatrix} A & C \\ C^\top & B \end{pmatrix}$$

is positive semidefinite.

Assume first that A and B are positive definite. Let

$$K := A^{-1/2} C B^{-1/2}.$$

By congruence,

$$\begin{pmatrix} I & K \\ K^\top & I \end{pmatrix} \succeq 0.$$

Hence $\|K\|_{\text{op}} \leq 1$. By duality between the operator norm and the nuclear norm,

$$\text{Tr } C = \text{Tr}(A^{1/2} K B^{1/2}) = \text{Tr}(K B^{1/2} A^{1/2}) \leq \|B^{1/2} A^{1/2}\|_*.$$

Moreover,

$$\|B^{1/2} A^{1/2}\|_* = \text{Tr} \left((A^{1/2} B A^{1/2})^{1/2} \right).$$

Thus the claim holds when A, B are positive definite.

For general $A, B \in \mathbb{S}_+^d$, apply the positive-definite case to $A + \varepsilon I$ and $B + \varepsilon I$, then let $\varepsilon \downarrow 0$. $\square$

Let $\pi \in \Pi(\mu, \nu)$, and let (U, V) have law π . Define

$$X := U - m_\mu, \quad Y := V - m_\nu.$$

Then

$$\text{Cov}(X) = \Sigma_\mu, \quad \text{Cov}(Y) = \Sigma_\nu.$$

Also,

$$\mathbb{E}\|U - V\|^2 = \|m_\mu - m_\nu\|^2 + \mathbb{E}\|X - Y\|^2.$$

Expanding,

$$\mathbb{E}\|X - Y\|^2 = \text{Tr } \Sigma_\mu + \text{Tr } \Sigma_\nu - 2 \text{Tr } \mathbb{E}[XY^\top].$$

By the lemma,

$$\text{Tr } \mathbb{E}[XY^\top] \leq \text{Tr} \left((\Sigma_\mu^{1/2} \Sigma_\nu \Sigma_\mu^{1/2})^{1/2} \right).$$

Therefore every coupling π satisfies

$$\int \|u - v\|^2 d\pi(u, v) \geq \|m_\mu - m_\nu\|^2 + \mathfrak{B}^2(\Sigma_\mu, \Sigma_\nu).$$

Taking the infimum over $\pi \in \Pi(\mu, \nu)$ gives the result. $\square$

Corollary 1. *The map $R : \mathcal{P}_2(\mathbb{R}^d) \rightarrow \mathcal{G}_d$ is 1-Lipschitz:*

$$W_2(R\mu, R\nu) \leq W_2(\mu, \nu).$$

Proof. For Gaussian laws,

$$W_2^2(\mathbf{N}(m, A), \mathbf{N}(n, B)) = \|m - n\|^2 + \mathfrak{B}^2(A, B).$$

Applying Gelbrich's inequality to μ and ν gives

$$W_2^2(R\mu, R\nu) = \|m_\mu - m_\nu\|^2 + \mathfrak{B}^2(\Sigma_\mu, \Sigma_\nu) \leq W_2^2(\mu, \nu).$$

□

3 A Gaussian-preservation criterion

The following criterion was communicated by Hugo Lavenant.

Proposition 1 (Variational Gaussian-preservation criterion). *Let $F : \mathcal{P}_2(\mathbb{R}^d) \rightarrow (-\infty, +\infty]$ satisfy*

$$F(R\mu) \leq F(\mu) \quad \text{for every } \mu \in \mathcal{P}_2(\mathbb{R}^d).$$

Fix $\tau > 0$, and consider one step of the JKO scheme:

$$\nu \in \operatorname{argmin}_{\eta \in \mathcal{P}_2(\mathbb{R}^d)} \left\{ F(\eta) + \frac{1}{2\tau} W_2^2(\gamma, \eta) \right\}.$$

If $\gamma \in \mathcal{G}_d$, then $R\nu$ is also a minimizer. In particular, if the minimizer is unique, then $\nu = R\nu$ and ν is Gaussian.

Consequently, whenever the Wasserstein gradient flow of F is unique and obtained as the limit of the JKO scheme, the flow preserves Gaussian initial data.

Proof. Since γ is Gaussian,

$$R\gamma = \gamma.$$

For any competitor η ,

$$F(R\eta) \leq F(\eta)$$

by assumption, and

$$W_2(R\gamma, R\eta) \leq W_2(\gamma, \eta)$$

by the previous corollary. Hence

$$F(R\eta) + \frac{1}{2\tau} W_2^2(\gamma, R\eta) \leq F(\eta) + \frac{1}{2\tau} W_2^2(\gamma, \eta).$$

Applying this to a minimizer $\eta = \nu$ shows that $R\nu$ is also a minimizer. If the minimizer is unique, then $\nu = R\nu$, hence ν is Gaussian. □

4 Examples satisfying the criterion

Example 1 (Functionals depending only on mean and covariance). *Let*

$$F(\mu) = \Phi(m_\mu, \Sigma_\mu)$$

for some function

$$\Phi : \mathbb{R}^d \times \mathbb{S}_+^d \rightarrow (-\infty, +\infty].$$

Then

$$F(R\mu) = F(\mu).$$

Thus the JKO scheme, when started from a Gaussian law, admits a Gaussian minimizer at each step.

In particular, quadratic interaction energies are covered. If

$$F(\mu) = \frac{1}{2} \int_{\mathbb{R}^d \times \mathbb{R}^d} (x - y)^\top Q (x - y) \, d\mu(x) \, d\mu(y),$$

with $Q = Q^\top$, then

$$F(\mu) = \text{Tr}(Q\Sigma_\mu).$$

Hence F depends only on Σ_μ , so $F(R\mu) = F(\mu)$.

Proof. The first claim is immediate from

$$m_{R\mu} = m_\mu, \quad \Sigma_{R\mu} = \Sigma_\mu.$$

For the quadratic interaction,

$$\mathbb{E}[(X - Y)^\top Q (X - Y)] = \text{Tr}\left(Q \mathbb{E}[(X - Y)(X - Y)^\top]\right),$$

where X, Y are independent with law μ . Since

$$\mathbb{E}[(X - Y)(X - Y)^\top] = 2\Sigma_\mu,$$

the prefactor $1/2$ gives

$$F(\mu) = \text{Tr}(Q\Sigma_\mu).$$

□

Example 2 (Boltzmann entropy). *Let*

$$\text{Ent}(\mu) := \begin{cases} \int_{\mathbb{R}^d} \rho \log \rho \, dx, & \mu = \rho \, dx, \\ +\infty, & \text{otherwise.} \end{cases}$$

Then

$$\text{Ent}(R\mu) \leq \text{Ent}(\mu).$$

Hence the entropy flow preserves Gaussian initial data.

Proof. Assume first that $\mu = \rho \, dx$ has finite entropy and nondegenerate covariance Σ_μ . Let

$$g$$

be the density of $R\mu = \mathcal{N}(m_\mu, \Sigma_\mu)$. Since relative entropy is nonnegative,

$$0 \leq \int_{\mathbb{R}^d} \rho \log \frac{\rho}{g} \, dx = \int_{\mathbb{R}^d} \rho \log \rho \, dx - \int_{\mathbb{R}^d} \rho \log g \, dx.$$

Now $\log g$ is a constant plus a quadratic polynomial:

$$\log g(x) = c - \frac{1}{2}(x - m_\mu)^\top \Sigma_\mu^{-1} (x - m_\mu).$$

Because μ and $R\mu$ have the same mean and covariance,

$$\int_{\mathbb{R}^d} \rho \log g \, dx = \int_{\mathbb{R}^d} g \log g \, dx = \text{Ent}(R\mu).$$

Therefore

$$\text{Ent}(\mu) \geq \text{Ent}(R\mu).$$

The singular case follows by the usual lower-semicontinuous extension, or by restricting to the affine span of the covariance. □

Example 3 (Squared Wasserstein distance to a Gaussian target). *Let $\gamma \in \mathcal{G}_d$ and define*

$$F(\mu) := \frac{1}{2}W_2^2(\mu, \gamma).$$

Then

$$F(R\mu) \leq F(\mu).$$

Proof. Since γ is Gaussian,

$$R\gamma = \gamma.$$

Therefore, by the 1-Lipschitz property of R ,

$$W_2(R\mu, \gamma) = W_2(R\mu, R\gamma) \leq W_2(\mu, \gamma).$$

Squaring gives $F(R\mu) \leq F(\mu)$. □

References

- [1] M. Gelbrich, On a formula for the L^2 -Wasserstein metric between measures on Euclidean and Hilbert spaces, *Mathematische Nachrichten* **147** (1990), 185–203.
- [2] H. Lavenant, Gaussian-preserving Wasserstein flows, personal communication, October 2025.